THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 1	The degree is the highest power of the variable.
2	**(d)** 11	Substitute x and simplify.
3	**(d)** 5	Solve $p(x) = 0$.
4	**(b)** 5	Put $x = 0$: $y = c$.
5	**(d)** 3	In $y = mx + c$ the slope is m.
6	**(b)** straight line	Recall the definition.
7	**(a)** $3n + 2$	$a_n = a_1 + (n-1)d$.
8	**(c)** $(0, 4)$	At $x = 0$, $y = c$.
9	**(b)** 3	The degree is the highest power of the variable.
10	**(c)** 14	Substitute x and simplify.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(2) = 3(2) + (5)$ (1); $p(2) = 11$ (1).	1+1
12	Put $3x - 15 = 0$ (1); $x = 5$ (1).	1+1
13	Degree $= 1$ (1); The polynomial is linear (1).	1+1
14	$p(5) = 4(5) - 3$ (1); $p(5) = 17$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x - 10 = 0$ gives $x = 5$ (1); Check: $p(5) = 2(5) + (-10) = 0$ (1); Hence the zero is verified (1).	1+1+1
16	$x = 0$: $y = 1$ (1); $x = 1$: $y = 4$ (1); $x = 2$: $y = 7$ (1).	1+1+1
17	$d = 4$ (1); $a_n = 4n - 1$ (1); $a_{10} = 39$ (1).	1+1+1
	OR: $3x + 12 = 0$ gives $x = -4$ (1); Check: $p(-4) = 3(-4) + (12) = 0$ (1); Hence the zero is verified (1).	1+1+1
18	$x = 0$: $y = 5$ (1); $x = 1$: $y = 7$ (1); $x = 2$: $y = 9$ (1).	1+1+1
	OR: $d = 5$ (1); $a_n = 5n + 1$ (1); $a_{12} = 61$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$F(x) = 10x + 30$ (1); $F(6) = 10 \times 6 + 30$ (1); $F(6) = ₹90$ (1); $10x + 30 = 110$ (1); $x = 8$ km (1).	1+1+1+1+1
	OR: Slope $= 2$ (1); y-intercept $= 6$ (1); x-intercept: $0 = 2x + (6)$, $x = -3$ (1); At $x = 3$: $y = 12$ (1); So the point lies on the line (1).	1+1+1+1+1
20	$F(x) = 12x + 20$ (1); $F(5) = 12 \times 5 + 20$ (1); $F(5) = ₹80$ (1); $12x + 20 = 104$ (1); $x = 7$ km (1).	1+1+1+1+1
	OR: Slope $= 3$ (1); y-intercept $= -9$ (1); x-intercept: $0 = 3x + (-9)$, $x = 3$ (1); At $x = 4$: $y = 3$ (1); The point has $y = 5 \neq 3$, so it does not lie on the line (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 3	Solve $p(x) = 0$.
2	**(d)** 7	Put $x = 0$: $y = c$.
3	**(b)** -4	In $y = mx + c$ the slope is m.
4	**(d)** one zero	Recall the definition.
5	**(a)** $2n + 2$	$a_n = a_1 + (n - 1)d$.
6	**(d)** $(0, -2)$	At $x = 0$, $y = c$.
7	**(a)** 2	The degree is the highest power of the variable.
8	**(d)** 10	Substitute x and simplify.
9	**(a)** 4	Solve $p(x) = 0$.
10	**(d)** -1	Put $x = 0$: $y = c$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Put $2x - 18 = 0$ (1); $x = 9$ (1).	1+1
12	Degree $= 0$ (1); The polynomial is constant (1).	1+1
13	$p(3) = -2(3) + (8)$ (1); $p(3) = 2$ (1).	1+1
14	Put $6x - 12 = 0$ (1); $x = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x - 20 = 0$ gives $x = 5$ (1); Check: $p(5) = 4(5) + (-20) = 0$ (1); Hence the zero is verified (1).	1+1+1
16	$x = 0$: $y = -2$ (1); $x = 1$: $y = 2$ (1); $x = 2$: $y = 6$ (1).	1+1+1
17	$d = 3$ (1); $a_n = 3n + 7$ (1); $a_{15} = 52$ (1).	1+1+1
	OR: $5x + 15 = 0$ gives $x = -3$ (1); Check: $p(-3) = 5(-3) + (15) = 0$ (1); Hence the zero is verified (1).	1+1+1
18	$x = 0$: $y = 0$ (1); $x = 1$: $y = 5$ (1); $x = 2$: $y = 10$ (1).	1+1+1
	OR: $d = 6$ (1); $a_n = 6n - 4$ (1); $a_{11} = 62$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$F(x) = 15x + 40$ (1); $F(4) = 15 \times 4 + 40$ (1); $F(4) =$ ₹100 (1); $15x + 40 = 175$ (1); $x = 9$ km (1).	1+1+1+1+1
	OR: Slope $= 4$ (1); y-intercept $= 8$ (1); x-intercept: $0 = 4x + (8)$, $x = -2$ (1); At $x = 2$: $y = 16$ (1); So the point lies on the line (1).	1+1+1+1+1
20	$F(x) = 8x + 25$ (1); $F(10) = 8 \times 10 + 25$ (1); $F(10) =$ ₹105 (1); $8x + 25 = 73$ (1); $x = 6$ km (1).	1+1+1+1+1
	OR: Slope $= 5$ (1); y-intercept $= -10$ (1); x-intercept: $0 = 5x + (-10)$, $x = 2$ (1); At $x = 5$: $y = 15$ (1); The point has $y = 18 \neq 15$, so it does not lie on the line (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
- Calling $x^2 - 5$ or $\frac{1}{x} + 2$ a linear polynomial; the degree must be exactly 1.
- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
- Reading the fixed charge as the slope, and the per-unit rate as the intercept, in word problems.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Linear Polynomials** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	In $y = mx + c$ the slope is m.
2	**(b)** constant polynomial	Recall the definition.
3	**(c)** $5n + 2$	$a_n = a_1 + (n-1)d$.
4	**(c)** $(0, 6)$	At $x = 0$, $y = c$.
5	**(c)** 0	The degree is the highest power of the variable.
6	**(b)** 3	Substitute x and simplify.
7	**(b)** 2	Solve $p(x) = 0$.
8	**(c)** 6	Put $x = 0$: $y = c$.
9	**(a)** 7	In $y = mx + c$ the slope is m.
10	**(a)** linear polynomial	Recall the definition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Degree $= 2$ (1); The polynomial is quadratic (1).	1+1
12	$p(4) = 7(4) + (1)$ (1); $p(4) = 29$ (1).	1+1
13	Put $5x - 20 = 0$ (1); $x = 4$ (1).	1+1
14	Degree $= 1$ (1); The polynomial is linear (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$7x - 21 = 0$ gives $x = 3$ (1); Check: $p(3) = 7(3) + (-21) = 0$ (1); Hence the zero is verified (1).	1+1+1
16	$x = 0$: $y = 7$ (1); $x = 1$: $y = 6$ (1); $x = 2$: $y = 5$ (1).	1+1+1
17	$d = 7$ (1); $a_n = 7n + 2$ (1); $a_{20} = 142$ (1).	1+1+1
	OR: $6x + 18 = 0$ gives $x = -3$ (1); Check: $p(-3) = 6(-3) + (18) = 0$ (1); Hence the zero is verified (1).	1+1+1
18	$x = 0$: $y = -4$ (1); $x = 1$: $y = 2$ (1); $x = 2$: $y = 8$ (1).	1+1+1
	OR: $d = 8$ (1); $a_n = 8n - 7$ (1); $a_9 = 65$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$F(x) = 20x + 50$ (1); $F(3) = 20 \times 3 + 50$ (1); $F(3) = ₹110$ (1); $20x + 50 = 150$ (1); $x = 5$ km (1).	1+1+1+1+1
	OR: Slope $= 6$ (1); y-intercept $= 12$ (1); x-intercept: $0 = 6x + (12)$, $x = -2$ (1); At $x = 1$: $y = 18$ (1); So the point lies on the line (1).	1+1+1+1+1
20	$F(x) = 9x + 15$ (1); $F(7) = 9 \times 7 + 15$ (1); $F(7) = ₹78$ (1); $9x + 15 = 123$ (1); $x = 12$ km (1).	1+1+1+1+1
	OR: Slope $= 2$ (1); y-intercept $= -8$ (1); x-intercept: $0 = 2x + (-8)$, $x = 4$ (1); At $x = 6$: $y = 4$ (1); The point has $y = 3 \neq 4$, so it does not lie on the line (1).	1+1+1+1+1

Mistakes to avoid:

- Writing the zero of $ax + b$ as $\frac{b}{a}$ instead of $-\frac{b}{a}$.
- Taking the slope of $y = ax + b$ to be b; the slope is the coefficient a and b is the y-intercept.
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- Finding the x-intercept by putting $x = 0$ (that gives the y-intercept); for the x-intercept put $y = 0$.
- In the nth term of a sequence, writing $t_n = dn + t_1$ instead of $dn + (t_1 - d)$, so that t_1 does not match.
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